Feasible normals and admissibility in the restricted bivariate normal problem under general correlation

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Setup and statement

Let

$$X \sim N_2(\mu, \Omega_\rho), \qquad \Omega_\rho = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \qquad -1 < \rho < 1 \ ,$$

and define

$$\Lambda = \{\mu : \mu_2 \leq \mu_1\} \ ,$$
$$\Lambda_0 = \{\mu \in \Lambda : \mu_1 \leq 0\} \ , \qquad \Lambda_1 = \{\mu \in \Lambda : \mu_1 > 0\} \ .$$

A nonrandomized test is written as

$$\phi_A(x) = I\{x \notin A\} \ ,$$

where $A \subseteq \mathbf{R}^2$ is the acceptance region. Throughout this note, A is assumed to be closed, convex, and coordinatewise lower:

$$x \in A, \quad y \leq x \quad \Longrightarrow \quad y \in A \ .$$

For a closed convex set C, write

$$h_C(a) := \sup_{x \in C} a'x$$

for its support function, allowing the value $+\infty$.

Relative to $P_0 = N_2(0, \Omega_\rho)$, the natural parameter corresponding to the mean μ is $\vartheta = \Omega_\rho^{-1}\mu$. A supporting normal a therefore generates the mean ray $\mu_\lambda = \lambda\Omega_\rho a$. This suggests the feasible normal cone

$$\begin{aligned} \mathcal{F}_\rho &:= \{a \in \mathbf{R}^2 : a \geq 0, \ \Omega_\rho a \in \Lambda\} \\ &= \{a \in \mathbf{R}^2 : 0 \leq a_2 \leq a_1\} = \mathrm{cone}\{(1,0),(1,1)\} \ . \end{aligned} \tag{1}$$

The second equality follows because

$$(\Omega_\rho a)_1 - (\Omega_\rho a)_2 = (1-\rho)(a_1 - a_2) \ .$$

Thus the feasible normal cone does not depend on ρ, even though the likelihood-ratio geometry does. The phrase "all finite normals lie in $\mathcal{F}_\rho$" means

$$a \neq 0, \quad h_A(a) < \infty \quad \Longrightarrow \quad a \in \mathcal{F}_\rho \ . \tag{2}$$

For a lower set, finite support already forces $a \geq 0$, so failure of (2) is equivalent to the existence of some $a \neq 0$ with

$$h_A(a) < \infty \ , \qquad 0 \leq a_1 < a_2 \ .$$

The main result to be established is the following characterization.

Theorem 1 (Feasible-normal characterization)**.** *Let $A \subseteq \mathbf{R}^2$ be closed, convex, and coordinatewise lower, and suppose*

$$P_0(X \in A) = 1 - \alpha \ , \qquad 0 < \alpha < 1 \ .$$

For every $-1 < \rho < 1$, the test $\phi_A = I\{X \notin A\}$ is α-admissible against all level-α tests if and only if all finite normals of A lie in $\mathcal{F}_\rho$, that is, if and only if (2) *holds.*

The proof has two parts. The sufficient direction is the Lehmann–Romano ray argument. The necessary direction is constructive. Under nonnegative correlation, a bad finite normal can be repaired by the same hull and diagonal calibration used in the identity-covariance problem. Under negative correlation, the proof first checks whether A is lower along the second extreme ray of the transfer order. If it is not, a conditional monotone rearrangement strictly improves the test. If it is, the resulting slope bound restores the gain-loss ordering needed for the diagonal repair.

1. Sufficiency

We use the following standard consequence of the Lehmann–Romano complete-class result for exponential families.

Proposition 1 (Lehmann–Romano ray criterion)**.** *Consider an exponential family with natural statistic T and natural parameter space Θ. Let Θ_H be the null parameter set and let Θ' be an alternative set. Let A_0 be a closed convex acceptance region for a nonrandomized test. Suppose that whenever*

$$A_0 \subseteq \{t : a't \le c\}$$

for some nonzero a, there exist $\vartheta^* \in \Theta$ and $\lambda_n \to \infty$ such that

$$\vartheta^* + \lambda_n a \in \Theta' \ .$$

Then the test with acceptance region A_0 is d-admissible. If, in addition, the test has exact size α attained at a finite null point, then it is α-admissible. These conclusions allow randomized competing tests.

Lemma 1 (Sufficiency of feasible normals)**.** *If A is closed, convex, coordinatewise lower, satisfies $P_0(A) = 1 - \alpha$, and satisfies* (2)*, then ϕ_A is α-admissible against all level-α tests.*

Proof. Relative to P_0, the normal experiment has the canonical representation

$$\frac{dP_\mu}{dP_0}(x) = \exp\left\{(\Omega_\rho^{-1}\mu)'x - \frac{1}{2}\mu'\Omega_\rho^{-1}\mu\right\} \ ,$$

with natural parameter $\vartheta = \Omega_\rho^{-1}\mu$ and statistic $T(x) = x$. The alternative set in the natural parameter space is

$$\Theta' = \Omega_\rho^{-1}\Lambda_1 \ .$$

We verify the ray condition in Proposition 1. Suppose

$$A \subseteq \{x : a'x \le c\}$$

for some nonzero a. Then $h_A(a) < \infty$, so by (2), $a \in \mathcal{F}_\rho$. Consider the natural-parameter ray $\vartheta_\lambda = \lambda a$. Its corresponding mean is

$$\mu_\lambda = \lambda \Omega_\rho a \ .$$

By (1), $\mu_\lambda \in \Lambda$. Moreover, $a \neq 0$ and $0 \leq a_2 \leq a_1$ imply $a_1 > 0$, and

$$(\mu_\lambda)_1 = \lambda(a_1 + \rho a_2) > 0 \ .$$

Indeed, this is immediate when $\rho \geq 0$, while for $\rho < 0$,

$$a_1 + \rho a_2 \geq (1 + \rho) a_1 > 0 \ .$$

Hence $\mu_\lambda \in \Lambda_1$, or equivalently $\vartheta_\lambda \in \Theta'$, for every $\lambda > 0$. The ray condition holds with $\vartheta^* = 0$ and any sequence $\lambda_n \to \infty$.

Because A is coordinatewise lower, the rejection region A^c is coordinatewise upper. For every $\mu \in \Lambda_0$, we have $\mu \leq 0$. Coupling $X = \mu + Z$ with $Z \sim N_2(0, \Omega_\rho)$ gives $\mu + Z \leq Z$ coordinatewise, and therefore

$$P_\mu(X \notin A) \leq P_0(Z \notin A) = \alpha \ .$$

Thus the test has level α, and exact size is attained at the finite null point $\mu = 0$. Proposition 1 gives α-admissibility against all level-α tests, including randomized competitors. □

2. The transfer order

The necessity proof uses the order with respect to which every restricted-alternative likelihood ratio is increasing. Define

$$T_{1,\rho}(x) := (\Omega_\rho^{-1}\mathbf{1})'x = \frac{x_1 + x_2}{1 + \rho} \ , \qquad T_{2,\rho}(x) := -(\Omega_\rho^{-1} e_2)'x = \frac{\rho x_1 - x_2}{1 - \rho^2} \ ,$$

and write

$$x \succeq_{\Lambda,\rho} y \quad \Longleftrightarrow \quad T_{1,\rho}(x) \geq T_{1,\rho}(y) \quad \text{and} \quad T_{2,\rho}(x) \geq T_{2,\rho}(y) \ .$$

The positive cone of this order is

$$\begin{aligned} \Pi_\rho &:= \{d : x + d \succeq_{\Lambda,\rho} x\} \\ &= \text{cone}\{(1, \rho), (1, -1)\} \\ &= \{d : d_1 + d_2 \geq 0, \ \rho d_1 - d_2 \geq 0\} \ . \end{aligned} \tag{3}$$

Intuitively, Π_ρ is the cone of universally power-improving moves in sample space for the restricted alternatives. The direction $(1, \rho)$ raises the common-shift score $T_{1,\rho}$ while leaving $T_{2,\rho}$ unchanged. The direction $(1, -1)$ raises the slack score $T_{2,\rho}$ while leaving $T_{1,\rho}$ unchanged.

For $\mu \in \Lambda_1$, write

$$u := \mu_1 > 0 \ , \qquad v := \mu_1 - \mu_2 \geq 0, \qquad \mu = u\mathbf{1} - v e_2 \ .$$

Then, relative to P_0,

$$L_\mu(x) := \frac{dP_\mu}{dP_0}(x) = \exp\left\{uT_{1,\rho}(x) + vT_{2,\rho}(x) - \frac{1}{2}\mu'\Omega_\rho^{-1}\mu\right\} \ . \tag{4}$$

Thus L_μ is strictly increasing in $T_{1,\rho}$ and weakly increasing in $T_{2,\rho}$.

We first replace A by its closed $-\Pi_\rho$-hull,

$$\widehat{A}_\rho := \mathrm{cl}(A - \Pi_\rho) \ .$$

Equivalently, $\widehat{A}_\rho$ is the smallest closed set containing A that is closed under moves opposite to the restricted likelihood-ratio order. This operation removes boundary pieces whose supporting normals point in infeasible directions. It may add too much acceptance probability, so the next step will restore the original size by cutting $\widehat{A}_\rho$ with a feasible diagonal halfspace.

For a rectangle, the hull operation retains the diagonal repair from the identity-covariance problem because $(1,-1)$ remains an extreme ray of Π_ρ. The other extreme ray changes from $(1,0)$ to $(1,\rho)$. When $\rho \geq 0$, its negative is already a coordinatewise downward move and adds nothing to a lower acceptance region. When $\rho < 0$, it contributes an additional likelihood-ratio-improving transfer. Lemma 2 treats both cases at once.

Lemma 2 (The feasible-normal hull)**.** *The set $\widehat{A}_\rho$ is closed, convex, coordinatewise lower, and Π_ρ-lower, meaning*

$$x \in \widehat{A}_\rho \ , \quad p \in \Pi_\rho \quad \Longrightarrow \quad x - p \in \widehat{A}_\rho \ .$$

Moreover, every nonzero finite-support direction of $\widehat{A}_\rho$ lies in $\mathcal{F}_\rho$. If A has a finite-support direction a with $a_2 > a_1$, then

$$A \subsetneq \widehat{A}_\rho \qquad \textit{and} \qquad P_0(\widehat{A}_\rho) > P_0(A) \ .$$

Proof. Because $\widehat{A}_\rho$ is the closure of the Minkowski sum $A + (-\Pi_\rho)$, it is closed and convex. It contains A because $0 \in \Pi_\rho$. It is coordinatewise lower because A is coordinatewise lower: if $x = y - p$ with $y \in A$ and $p \in \Pi_\rho$, and if $z \leq x$, then

$$z = (y - (x - z)) - p \ ,$$

where $y - (x - z) \in A$. Passing to the closure preserves the lower-set property. The Π_ρ-lower property follows from the definition.

The support function satisfies

$$h_{\widehat{A}_\rho}(a) = h_A(a) + h_{-\Pi_\rho}(a) \ ,$$

with the usual convention for extended-real sums. Since $-\Pi_\rho$ is generated by $(-1,-\rho)$ and $(-1,1)$,

$$h_{-\Pi_\rho}(a) < \infty \quad \Longleftrightarrow \quad a_1 + \rho a_2 \geq 0 \quad \text{and} \quad a_2 \leq a_1 \ . \tag{5}$$

Also, since A is coordinatewise lower and nonempty, $h_A(a) < \infty$ implies $a \geq 0$. If $h_{\widehat{A}_\rho}(a) < \infty$ and $a \neq 0$, it follows that $a \geq 0$ and $a_2 \leq a_1$. The remaining inequality in (5) is then automatic: it is immediate for $\rho \geq 0$, while for $\rho < 0$,

$$a_1 + \rho a_2 \geq (1+\rho)a_1 \geq 0 \ .$$

Therefore

$$0 \leq a_2 \leq a_1 \ ,$$

so $a \in \mathcal{F}_\rho$.

Finally, suppose that $h_A(a) < \infty$ for some a with $a_2 > a_1$. Then $h_{\widehat{A}_\rho}(a) = +\infty$, because a is not in the polar cone of $-\Pi_\rho$. Indeed, $(-1, 1) \in -\Pi_\rho$ and

$$a'(-1,1) = a_2 - a_1 > 0 \ .$$

Every point $x \in A$ therefore generates the ray $x + t(-1,1) \in A - \Pi_\rho$. Along this ray the support functional tends to $+\infty$:

$$a'\{x + t(-1,1)\} = a'x + t(a_2 - a_1) \longrightarrow +\infty \ .$$

Hence $\widehat{A}_\rho$ is not contained in the supporting halfspace $\{x : a'x \leq h_A(a)\}$ that contains A, so $\widehat{A}_\rho \setminus A$ is nonempty. Since $P_0(A) \in (0,1)$ and A is convex, A has nonempty interior. Taking a point of $\widehat{A}_\rho$ outside that supporting halfspace and joining it to an interior point of A gives points of $\mathrm{int}(\widehat{A}_\rho) \setminus A$. Thus $\widehat{A}_\rho \setminus A$ contains a nonempty open set. The density of $N_2(0, \Omega_\rho)$ is everywhere positive, and therefore

$$P_0(\widehat{A}_\rho) > P_0(A) \ .$$

□

Lemma 3 (Calibration of the competitor)**.** *Suppose that $P_0(A) = 1 - \alpha$ and that A has a finite-support direction a satisfying $a_2 > a_1$. Then there exists a finite number b such that*

$$P_0\big(\widehat{A}_\rho \cap \{x : T_{1,\rho}(x) \leq b\}\big) = 1 - \alpha \ .$$

For any such b, define

$$A' := \widehat{A}_\rho \cap \{x : T_{1,\rho}(x) \leq b\} \ .$$

Then A' is closed, convex, and coordinatewise lower, satisfies $P_0(A') = 1 - \alpha$, and every nonzero finite-support direction of A' lies in $\mathcal{F}_\rho$. Consequently, the test $\phi_{A'}(x) = I\{x \notin A'\}$ has level α, attained at $\mu = 0$.

Proof. By Lemma 2,

$$P_0(\widehat{A}_\rho) > P_0(A) = 1 - \alpha \ .$$

The map

$$t \longmapsto P_0\big(\widehat{A}_\rho \cap \{x : T_{1,\rho}(x) \leq t\}\big)$$

is continuous because $T_{1,\rho}(X)$ has a nondegenerate normal distribution under P_0. It converges to zero as $t \to -\infty$ and to $P_0(\widehat{A}_\rho)$ as $t \to \infty$. Hence there exists a finite b for which the displayed probability equals $1 - \alpha$.

The set A' is closed and convex because it is the intersection of two sets with these properties. It is also coordinatewise lower. Indeed,

$$T_{1,\rho}(x) = \frac{x_1 + x_2}{1 + \rho}$$

is coordinatewise increasing because $1 + \rho > 0$. Its null probability equals $1 - \alpha$ by construction.

It remains to characterize the finite-support directions of A'. We first show that A' is Π_ρ-lower. Let $x \in A'$ and $p \in \Pi_\rho$. Lemma 2 implies that $x - p \in \widehat{A}_\rho$. Moreover, $T_{1,\rho}(p) \geq 0$ by the definition of Π_ρ, and therefore

$$T_{1,\rho}(x - p) = T_{1,\rho}(x) - T_{1,\rho}(p) \leq T_{1,\rho}(x) \leq b \ .$$

Thus $x - p \in A'$.

Now let $a \neq 0$ satisfy $h_{A'}(a) < \infty$, and fix $x \in A'$. Since A' is coordinatewise lower, $x - te_j \in A'$ for every $t \geq 0$ and $j = 1, 2$. If $a_j < 0$, then

$$a'(x - te_j) = a'x - ta_j \longrightarrow +\infty \ ,$$

contradicting $h_{A'}(a) < \infty$. Hence $a_1 \geq 0$ and $a_2 \geq 0$.

Because $(1, -1) \in \Pi_\rho$ and A' is Π_ρ-lower,

$$x - t(1, -1) \in A' \qquad \text{for every } t \geq 0 \ .$$

If $a_2 > a_1$, then

$$a'\{x - t(1, -1)\} = a'x + t(a_2 - a_1) \longrightarrow +\infty \ ,$$

again contradicting finite support. Therefore $a_2 \leq a_1$. We have shown that every nonzero a with $h_{A'}(a) < \infty$ satisfies

$$0 \leq a_2 \leq a_1 \ ,$$

so $a \in \mathcal{F}_\rho$.

Finally, the complement of A' is coordinatewise upper. For every $\mu \in \Lambda_0$, we have $\mu \leq 0$. Coupling $X = \mu + Z$ with $Z \sim N_2(0, \Omega_\rho)$ gives

$$P_\mu(X \notin A') \leq P_0(Z \notin A') = \alpha \ .$$

Equality holds at the finite null point $\mu = 0$, which proves the last claim. □

Lemma 4 (Transfer domination criterion)**.** *Let G and D be Borel sets with*

$$P_0(G) = P_0(D) > 0 \ .$$

Suppose that, for every $g \in G$ and $d \in D$,

$$T_{1,\rho}(g) > T_{1,\rho}(d) \qquad \textit{and} \qquad T_{2,\rho}(g) \geq T_{2,\rho}(d) \ .$$

Then

$$P_\mu(G) > P_\mu(D) \qquad \textit{for every } \mu \in \Lambda_1 \ .$$

Proof. Let $m = P_0(G) = P_0(D)$. Using the likelihood ratio in (4),

$$m\{P_\mu(G) - P_\mu(D)\} = \int_G \int_D \{L_\mu(g) - L_\mu(d)\}\, dP_0(d)\, dP_0(g) \ .$$

For $\mu \in \Lambda_1$, the coefficient on $T_{1,\rho}$ in (4) is $u = \mu_1 > 0$, while the coefficient on $T_{2,\rho}$ is $v = \mu_1 - \mu_2 \geq 0$. The assumed ordering therefore gives

$$L_\mu(g) > L_\mu(d)$$

for every pair $(g, d) \in G \times D$. Since $m > 0$, the double integral is strictly positive, which proves the result. □

3. Necessity

It remains to prove that a bad finite normal generates sets G and D satisfying Lemma 4. The proof is slicewise. The first step is to observe that, although the transfer cone Π_ρ depends on ρ, its lower hull does not. Let

$$K := \mathrm{cone}\{(1, -1)\} \ .$$

Since $K \subseteq \Pi_\rho$, we have $A - K \subseteq A - \Pi_\rho$. For the reverse inclusion, recall that

$$\Pi_\rho = \mathrm{cone}\{(1, \rho), (1, -1)\} \ .$$

If $\rho \geq 0$, subtracting a nonnegative multiple of $(1, \rho)$ is a coordinatewise downward move and is already absorbed by A. If $\rho < 0$, then

$$(1, \rho) = (-\rho)(1, -1) + (1 + \rho)e_1 \ ,$$

where both coefficients are positive. Subtracting the second term is again a coordinatewise downward move. Hence, in both cases,

$$A - \Pi_\rho = A - K, \qquad \widehat{A}_\rho = \mathrm{cl}(A - K) \ . \tag{6}$$

For $s \in \mathbf{R}$, define the horizontal boundary of A by

$$\beta(s) := \sup\{x_1 : (x_1, s) \in A\} \in [-\infty, +\infty] \ .$$

Since A is closed, convex, and coordinatewise lower,

$$A \cap \{x_2 = s\} = \{(x_1, s) : x_1 \leq \beta(s)\} \ ,$$

with the usual interpretation when $\beta(s) = \pm\infty$. The function β is concave in the extended-real sense. Set

$$m(s) := \beta(s) + s, \qquad M(s) := \sup_{r \le s} m(r) \ .$$

Thus $m(s)$ is the largest value of the unscaled diagonal score $x_1 + x_2 = (1+\rho)T_{1,\rho}(x)$ on the horizontal slice of A, and $M(s)$ is its running maximum from below.

Lemma 5 (Slice formula for $\widehat{A}_\rho$)**.** *For every $s \in \mathbf{R}$,*

$$\widehat{A}_\rho \cap \{x_2 = s\} = \{(x_1, s) : x_1 \le M(s) - s\} \ .$$

Equivalently,

$$(x_1, s) \in \widehat{A}_\rho \quad \Longleftrightarrow \quad x_1 + s \le M(s) \ .$$

Proof. We first verify that the set

$$C := \{(x_1, s) : x_1 + s \le M(s)\}$$

is closed. The function M is never $-\infty$. To see this, fix $(x_1^0, s_0) \in A$. If $s \ge s_0$, then $M(s) \ge m(s_0) > -\infty$; if $s < s_0$, coordinatewise lowerness gives $(x_1^0, s) \in A$, so $M(s) \ge m(s) > -\infty$.

The function M is nondecreasing. It is also concave whenever it is finite. Indeed, for $s_1, s_2 \in \mathbf{R}$, choose $r_i \le s_i$ such that $m(r_i) \ge M(s_i) - \varepsilon$, $i = 1, 2$. For every $\lambda \in [0, 1]$, concavity of m gives

$$\begin{aligned} M\{\lambda s_1 + (1-\lambda)s_2\} &\ge m\{\lambda r_1 + (1-\lambda)r_2\} \\ &\ge \lambda M(s_1) + (1-\lambda)M(s_2) - \varepsilon \ . \end{aligned}$$

Letting $\varepsilon \downarrow 0$ proves the claim.

If M takes the value $+\infty$ at some point, then it is identically $+\infty$. Monotonicity gives this conclusion to the right. To see it to the left, suppose $M(s_0) = +\infty$ and fix $t < s_0$. Choose $r_n \le s_0$ with $m(r_n) \to +\infty$. If $r_n \le t$ along a subsequence, then $M(t) = +\infty$. Otherwise, $t < r_n \le s_0$ eventually. Since A is coordinatewise lower, $\beta(t) \ge \beta(r_n)$, and hence

$$m(t) \ge m(r_n) + t - r_n \ge m(r_n) + t - s_0 \longrightarrow +\infty \ .$$

Thus $M(t) = +\infty$ in this case as well. It follows that either $M \equiv +\infty$, in which case $C = \mathbf{R}^2$, or M is a finite concave function on $\mathbf{R}$. In the latter case M is continuous. Therefore C is closed in either case.

Now suppose $(x_1, s) \in A - K$. By (6), there exist $(y_1, r) \in A$ and $\tau \ge 0$ such that

$$(x_1, s) = (y_1, r) - \tau(1, -1) \ .$$

Hence $r = s - \tau \le s$ and

$$x_1 + s = y_1 + r \le m(r) \le M(s) \ .$$

Thus $A - K \subseteq C$. Since C is closed, $\widehat{A}_\rho = \mathrm{cl}(A - K) \subseteq C$.

Conversely, suppose $x_1 + s < M(s)$. Then there exists $r \le s$ such that

$$m(r) > x_1 + s \ .$$

Choose $y_1 \le \beta(r)$ with $y_1 + r > x_1 + s$. Then $(y_1, r) \in A$. Since A is coordinatewise lower, the point

$$\widetilde{y}_1 := x_1 + s - r < y_1$$

also satisfies $(\widetilde{y}_1, r) \in A$. Setting $\tau = s - r \ge 0$ gives

$$(x_1, s) = (\widetilde{y}_1, r) - \tau(1, -1) \in A - K \ .$$

Thus every point satisfying the strict inequality belongs to $A - K$. If $x_1 + s = M(s)$, apply this argument to $(x_1 - 1/n, s)$ and let $n \to \infty$. Since $\widehat{A}_\rho$ is closed, the point (x_1, s) also belongs to $\widehat{A}_\rho$. This proves the formula. □

For $\rho < 0$, write

$$q_\rho := (1, \rho) \ ,$$

and say that a set C is q_ρ-lower if

$$x \in C, \quad t \ge 0 \quad \Longrightarrow \quad x - tq_\rho \in C \ .$$

This property is automatic when $\rho \ge 0$ and C is coordinatewise lower, but not when $\rho < 0$: subtracting q_ρ lowers the first coordinate and raises the second. The next two lemmas handle the two possibilities.

Lemma 6 (Conditional rearrangement)**.** *Suppose $\rho < 0$. If A is not q_ρ-lower, then there exists a Borel acceptance region A^{r} such that the test $\phi_{A^{\mathrm{r}}} = I\{X \notin A^{\mathrm{r}}\}$ has level α and*

$$P_\mu(A^{\mathrm{r}}) < P_\mu(A) \qquad \textit{for every } \mu \in \Lambda_1 \ .$$

Consequently, $\phi_{A^{\mathrm{r}}}$ strictly dominates ϕ_A.

Proof. Introduce the coordinates

$$Z = X_1, \qquad E = X_2 - \rho X_1 \ .$$

Writing $\mu = (u, u - v)$, where $v \ge 0$ on Λ, we have

$$Z \sim N(u, 1), \qquad E \sim N((1-\rho)u - v, 1 - \rho^2), \qquad Z \perp E \ .$$

For $e \in \mathbf{R}$, let

$$A_e := \{z : (z, e + \rho z) \in A\}, \qquad p(e) := \Phi(A_e),$$

where Φ denotes standard normal probability. By Fubini's theorem, p is measurable. Moreover, if $e_1 \le e_2$, coordinatewise lowerness of A gives $A_{e_2} \subseteq A_{e_1}$. Hence p is nonincreasing.

Let

$$c(e) := \Phi^{-1}\{p(e)\},$$

with the conventions $c(e) = -\infty$ when $p(e) = 0$ and $c(e) = +\infty$ when $p(e) = 1$, and define

$$A^{\mathrm{r}} := \{x : x_1 \le c(x_2 - \rho x_1)\} \ . \tag{7}$$

The function c is measurable and nonincreasing. Thus, in (z, e) coordinates, $A^{\mathrm{r}} = \{(z, e) : z \le c(e)\}$ is Borel and coordinatewise lower.

Under P_0, Z and E are independent and the conditional standard normal mass of every section is preserved. It follows that

$$P_0(A^{\mathrm{r}}) = P_0(A) = 1 - \alpha \ .$$

If $\mu \in \Lambda_0$, then $u \le 0$, $v \ge 0$, and

$$\big(u, (1-\rho)u - v\big) \le (0, 0) \ .$$

Since A^{r} is lower in (z, e) coordinates, translation coupling therefore gives

$$P_\mu(A^{\mathrm{r}}) \ge P_0(A^{\mathrm{r}}) = 1 - \alpha \ .$$

The rearranged test consequently has level α, with equality at $\mu = 0$.

Now fix $\mu \in \Lambda_1$, so that $u > 0$. Conditional on $E = e$, the likelihood ratio of $N(u, 1)$ relative to $N(0, 1)$ is strictly increasing in z. Among all subsets of the real line having standard normal probability $p(e)$, the left halfline $(-\infty, c(e)]$ therefore minimizes their $N(u, 1)$ probability. Applying this conditional Neyman–Pearson inequality and integrating over E yields

$$P_\mu(A^{\mathrm{r}}) \le P_\mu(A) \ . \tag{8}$$

It remains to prove strictness. Because A is not q_ρ-lower, there are $x \in A$ and $t > 0$ for which $x - t q_\rho \notin A$. The set A has nonempty interior. By moving x slightly toward an interior point of A, if necessary, we may find $x^0 \in \mathrm{int}(A)$ and $t_0 > 0$ such that $x^0 - t_0 q_\rho \notin A$. Closedness of A then gives small open neighborhoods of these two points, the first contained in A and the second contained in its complement. Under the invertible map

$$H(x) = (x_1, x_2 - \rho x_1),$$

the two centers have the same second coordinate and their first coordinates differ by t_0. The images of the neighborhoods consequently contain two rectangles with a common nonempty interval of e values. For every e in a smaller such interval, A_e contains an interval lying strictly to the right of an interval it omits. Thus A_e is not a left halfline modulo standard normal measure. The conditional Neyman–Pearson inequality is strict for all these e. Since the density of E is everywhere positive, integration makes (8) strict for every $u > 0$, and hence for every $\mu \in \Lambda_1$. □

Lemma 7 (Gain-loss separation)**.** *Let A' be defined as in Lemma 3, and define the gain and loss sets*

$$G := A \setminus A' = A \cap \{T_{1,\rho} > b\}, \qquad D := A' \setminus A \ .$$

Suppose either that $0 \le \rho < 1$, *or that* $-1 < \rho < 0$ *and* A *is* q_ρ*-lower. Then*

$$P_0(G) = P_0(D) > 0 \ .$$

Moreover, for every $g \in G$ *and every* $d \in D$,

$$T_{1,\rho}(g) > T_{1,\rho}(d) \qquad \textit{and} \qquad T_{2,\rho}(g) \ge T_{2,\rho}(d) \ .$$

Proof. Since $A \subseteq \widehat{A}_\rho$ and $A' = \widehat{A}_\rho \cap \{T_{1,\rho} \le b\}$,

$$A \setminus A' = A \cap \{T_{1,\rho} > b\} = G \ .$$

Also $D = A' \setminus A$. Because $P_0(A) = P_0(A') = 1 - \alpha$,

$$P_0(G) = P_0(D) \ .$$

This common mass is positive. Otherwise A and A' would be equal up to a Gaussian null set. Since both sets are closed, convex, and have positive Gaussian probability, they have nonempty interiors. If the sets were not equal, joining a point in one set but not the other to an interior point of the first set would produce a nonempty open subset of their symmetric difference. The Gaussian density is everywhere positive, so the symmetric difference could not have probability zero. Hence null symmetric difference would imply $A = A'$. This is impossible because Lemma 3 shows that every finite-support direction of A' lies in $\mathcal{F}_\rho$, whereas A has a finite-support direction outside $\mathcal{F}_\rho$.

We now prove the ordering. Write

$$B_\rho := (1+\rho)b \ ,$$

so that $T_{1,\rho}(x) \le b$ is equivalent to $x_1 + x_2 \le B_\rho$. On the slice $x_2 = s$, the gain set is

$$G_s := \{x_1 : (x_1, s) \in G\} = (B_\rho - s, \beta(s)] \ .$$

Thus

$$G_s \neq \varnothing \quad \Longleftrightarrow \quad m(s) > B_\rho \ . \tag{9}$$

Using Lemma 5, the loss slice is

$$D_s := \{x_1 : (x_1, s) \in D\} = (\beta(s), \min\{M(s), B_\rho\} - s] \ .$$

Consequently,

$$D_s \neq \varnothing \quad \Longleftrightarrow \quad M(s) > m(s) \quad \text{and} \quad m(s) < B_\rho \ . \tag{10}$$

Let

$$s_+ := \sup\{s : m(s) > B_\rho\} \ .$$

The set in braces is nonempty because $P_0(G) > 0$. We claim that

$$G_s \neq \varnothing \quad \Longrightarrow \quad s \le s_+,$$

and

$$D_s \neq \varnothing \quad \Longrightarrow \quad s \geq s_+ \ .$$

The first implication follows from the definition of s_+ and (9). For the second, suppose toward a contradiction that $D_s \neq \varnothing$ for some $s < s_+$. By (10), $M(s) > m(s)$, so there is $q \leq s$ such that

$$m(q) > m(s) \ .$$

Necessarily $q < s$. Since $s < s_+$, there is also $r > s$ such that

$$m(r) > B_\rho > m(s) \ .$$

Because $q < s < r$ and m is concave, for the unique $\lambda \in (0,1)$ satisfying $s = \lambda q + (1-\lambda)r$ we obtain

$$m(s) \geq \lambda m(q) + (1-\lambda) m(r) > m(s) \ ,$$

a contradiction. Hence no loss slice lies below s_+. Since D is nonempty, this also shows that s_+ is finite.

It follows that every $g \in G$ and $d \in D$ satisfy

$$g_2 \leq s_+ \leq d_2 \ .$$

In fact, $g_2 < d_2$, because a gain slice requires $m(s) > B_\rho$, whereas a loss slice requires $m(s) < B_\rho$. By definition of the gain and loss sets,

$$T_{1,\rho}(g) > b \geq T_{1,\rho}(d) \ .$$

If $\rho \geq 0$, the identity

$$T_{2,\rho}(x) = \frac{\rho T_{1,\rho}(x) - x_2}{1-\rho}$$

gives

$$T_{2,\rho}(g) - T_{2,\rho}(d) = \frac{\rho\{T_{1,\rho}(g) - T_{1,\rho}(d)\} + d_2 - g_2}{1-\rho} \geq 0 \ ,$$

which proves the result in this case.

Suppose instead that $\rho < 0$ and A is q_ρ-lower. If $s < t$ and $(x_1, s) \in A$, set $\tau = (t-s)/(-\rho)$. Then

$$(x_1, s) - \tau q_\rho = (x_1 - \tau, t) \in A \ .$$

It follows that

$$\beta(t) \geq \beta(s) - \frac{t-s}{-\rho}$$

whenever the relevant boundaries are finite, and hence

$$m(s) - m(t) \leq \frac{1+\rho}{-\rho}(t-s) \ . \tag{11}$$

For $g \in G$ and $d \in D$, put $s = g_2$ and $t = d_2$. We have $s < t$, and the boundaries at s and t are finite. Indeed, a gain point makes the slice at s nonempty, while q_ρ-lowerness carries

such a point to height t; moreover, $\beta(s) = +\infty$ would force $\beta(t) = +\infty$, contrary to the existence of a loss point at t. Therefore

$$g_1 + s \le m(s), \qquad d_1 + t > m(t),$$

and therefore, by (11),

$$(g_1 + g_2) - (d_1 + d_2) < \frac{1+\rho}{-\rho}(d_2 - g_2) \ .$$

Since $\rho < 0$, multiplication reverses this strict inequality. Consequently,

$$\begin{aligned}(1-\rho^2)\{T_{2,\rho}(g) - T_{2,\rho}(d)\} &= \rho\{(g_1+g_2) - (d_1+d_2)\} + (1+\rho)(d_2 - g_2) \\ &> 0 \ .\end{aligned}$$

This proves the asserted pairwise ordering in both cases. □

Lemma 8 (Necessity of feasible normals)**.** *If A has a finite-support direction a with $a_2 > a_1$, then ϕ_A is inadmissible. More precisely, there exists a level-α test ϕ^* such that*

$$P_\mu(\phi^* = 1) > P_\mu(\phi_A = 1) \qquad \textit{for every } \mu \in \Lambda_1 \ .$$

Proof. First suppose $\rho \ge 0$, and construct A' as in Lemma 3. That lemma shows that $\phi_{A'}$ has level α. Writing $G = A \setminus A'$ and $D = A' \setminus A$, the difference in power is

$$P_\mu(X \notin A') - P_\mu(X \notin A) = P_\mu(G) - P_\mu(D) \ .$$

Lemma 7 verifies the hypotheses of the transfer criterion, Lemma 4. Hence this difference is strictly positive for every $\mu \in \Lambda_1$.

Now suppose $\rho < 0$. If A is not q_ρ-lower, Lemma 6 supplies a level-α test that strictly dominates ϕ_A at every $\mu \in \Lambda_1$. If A is q_ρ-lower, construct A' as in Lemma 3. The negative-correlation part of Lemma 7 again verifies the hypotheses of Lemma 4, so $\phi_{A'}$ strictly dominates ϕ_A on Λ_1. These cases exhaust $-1 < \rho < 1$. □

Proof of Theorem 1. If all finite normals of A lie in $\mathcal{F}_\rho$, Lemma 1 gives α-admissibility against all level-α tests.

Conversely, suppose the finite-normal condition fails. Since A is coordinatewise lower, every finite-support direction is nonnegative. Thus there exists $a \ne 0$ such that

$$h_A(a) < \infty, \qquad 0 \le a_1 < a_2 \ .$$

Lemma 8 constructs a level-α test that strictly dominates ϕ_A at every $\mu \in \Lambda_1$. Therefore ϕ_A is inadmissible. □

Remark 1 (Why the identity case needs no extra step). When $\rho = 0$, the score $T_{2,0}(x) = -x_2$. The vertical separation $g_2 < d_2$ therefore gives the required score ordering directly. Equivalently, $q_0 = (1, 0)$, and coordinatewise lowerness already implies q_0-lowerness. For $\rho < 0$, neither fact remains true: $T_{2,\rho}$ contains a negative multiple of $T_{1,\rho}$, and subtracting q_ρ raises the second coordinate. The rearrangement lemma handles acceptance regions that fail this new directional monotonicity; the slope bound in Lemma 7 handles those that satisfy it. This is precisely the additional work required beyond the identity-covariance proof.